INSTRUMENTATION AND MONITORING SUMMARY
2015 – 2025

Elk Basin Dam
FERC Project No. 4182

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with
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1.0 INSTRUMENTATION OVERVIEW

Elk Basin Dam has been monitored since first filling in 1968-1969 using a comprehensive instrumentation system installed during construction. This summary presents 10 years of monitoring data (2015-2025) from the key instruments, with emphasis on the three conditions identified during the 2020 FERC Part 12D inspection: (1) left abutment seepage, (2) right abutment displacement, and (3) concrete arch behavior.

Table 1-1: Instrumentation Inventory

System	Qty Installed	Qty Functional	Frequency	Key Parameters
Seepage weirs	4	4	Monthly	Flow rate (gpm) at each abutment toe
Foundation piezometers	12	10	Monthly	Pore pressure (psi), head (ft)
Embankment piezometers	8	7	Monthly	Phreatic surface elevation
Survey monuments	16	16	Semi-annual	Horizontal and vertical displacement
Plumblines	3	3	Quarterly	Arch crest deflection (x, y)
Joint meters	8	6	Quarterly	Joint opening/closing (mm)
Concrete strain meters	24	16	Quarterly	Microstrain (locked-in + thermal)
Thermometers	18	14	Monthly	Concrete and air temperature
Reservoir level	1	1	Daily	Pool elevation (ft)
New VW piezometers (2025)	2	2	Monthly	BH-LA-01 (95 ft), BH-RA-01 (75 ft)
New inclinometer (2025)	1	1	Quarterly	BH-RA-01 (to 90 ft)

2.0 SEEPAGE MONITORING

Seepage is measured at four V-notch weirs located at the toe of the left and right abutment embankments and at two gallery drain outlets within the arch structure. The following table presents annual readings at consistent reservoir levels (near full pool, elevation 6,238 to 6,240 feet).

Table 2-1: Annual Seepage Summary (September readings, near full pool)

Year	Left Abt. Weir 1 (gpm)	Left Abt. Weir 2 (gpm)	Left Abt. Total (gpm)	Right Abt. Total (gpm)	Arch Gallery (gpm)	Total Dam (gpm)	Reservoir Elev. (ft)
2015	12.5	5.5	18.0	7.2	2.1	27.3	6,240.1
2016	13.2	5.8	19.0	7.0	2.2	28.2	6,239.5
2017	14.0	6.2	20.2	7.1	2.0	29.3	6,240.0
2018	15.5	7.0	22.5	7.3	2.1	31.9	6,239.2
2019	15.8	7.2	23.0	6.8	2.3	32.1	6,240.3
2020	18.0	10.0	28.0	7.0	2.0	37.0	6,240.0
2021	18.5	10.2	28.7	7.2	2.2	38.1	6,239.8

Year	Left Abt. Weir 1 (gpm)	Left Abt. Weir 2 (gpm)	Left Abt. Total (gpm)	Right Abt. Total (gpm)	Arch Gallery (gpm)	Total Dam (gpm)	Reservoir Elev. (ft)
2022	19.2	10.8	30.0	7.0	2.1	39.1	6,240.2
2023	19.8	11.5	31.3	7.3	2.0	40.6	6,239.6
2024	20.5	11.8	32.3	7.1	2.2	41.6	6,240.1
2025	21.2	12.5	33.7	7.2	2.1	43.0	6,239.9

Key observations: Left abutment seepage has increased 87 percent over 10 years (18.0 to 33.7 gpm), with both weirs showing consistent increases. The rate of increase has been approximately 1.5 gpm per year. Right abutment and arch gallery seepage remain stable. All seepage water remains clear with no turbidity, as confirmed by weekly visual inspections and quarterly turbidity meter readings (all below 1.0 NTU).

3.0 PIEZOMETRIC DATA

3.1 Foundation Piezometers — Left Abutment

Table 3-1: Left Abutment Foundation Piezometer Readings (September)

Year	P-1 (psi)	P-2 (psi)	P-3 (psi)	P-4 (psi)	P-5 (psi)	Reservoir Elev.
2015	6.8	5.2	10.5	7.8	4.2	6,240.1
2017	7.0	5.4	11.8	8.0	4.3	6,240.0
2019	7.2	5.5	13.2	8.2	4.5	6,240.3
2021	7.1	5.6	14.8	8.5	4.4	6,239.8
2023	7.3	5.5	16.2	8.8	4.6	6,239.6
2025	7.4	5.7	18.0	9.2	4.5	6,239.9

P-3 is the critical instrument. Located at Elevation 6,185 in the foundation below the left abutment clay core, P-3 shows a 71 percent increase (10.5 to 18.0 psi) over 10 years, closely correlating with the seepage increase. P-1, P-2, P-4, and P-5 show only minor fluctuations (less than 15 percent change), indicating the elevated pressures are localized to the P-3 vicinity — consistent with seepage through the weathered granodiorite zone identified by the 2025 boring program.

3.2 Embankment Piezometers — Right Abutment

Table 3-2: Right Abutment Embankment Piezometer Readings

Year	EP-6 (ft elev.)	EP-7 (ft elev.)	EP-8 (ft elev.)	Notes
2015	6,178.5	6,172.2	6,165.8	Stable
2017	6,178.8	6,172.0	6,165.5	Minor fluctuation
2019	6,178.2	6,172.4	6,166.0	Stable
2021	6,179.0	6,172.1	6,165.8	Stable
2023	6,178.5	6,172.5	6,166.2	Stable
2025	6,178.8	6,172.3	6,165.5	No trend detected

Right abutment embankment piezometers show stable pore pressures with less than 1.0 foot of variation over 10 years. This confirms that the observed crest displacement is not driven by changing pore pressure conditions, supporting the creep mechanism identified in the 2025 investigation.

4.0 SURVEY MONUMENT DISPLACEMENT

4.1 Right Abutment Crest Displacement

Table 4-1: Right Abutment Crest Displacement (Cumulative from 1995 Baseline)

Year	RA-1 DS (in)	RA-2 DS (in)	RA-3 DS (in)	RA-4 DS (in)	Avg DS (in)	Avg Vert (in)	Rate (in/yr)
2015	2.02	1.88	2.35	1.78	2.01	-0.32	0.19
2016	2.12	1.98	2.48	1.88	2.12	-0.34	0.20
2017	2.22	2.06	2.58	1.95	2.20	-0.35	0.19
2018	2.32	2.16	2.70	2.05	2.31	-0.37	0.20
2019	2.41	2.24	2.80	2.12	2.39	-0.38	0.19
2020	2.58	2.42	3.02	2.30	2.58	-0.41	0.20
2021	2.68	2.50	3.12	2.38	2.67	-0.42	0.20
2022	2.78	2.60	3.24	2.48	2.78	-0.43	0.20
2023	2.88	2.70	3.35	2.58	2.88	-0.45	0.19
2024	2.98	2.80	3.48	2.68	2.99	-0.46	0.20
2025	3.08	2.88	3.58	2.75	3.07	-0.48	0.19

Key findings: The displacement rate remains constant at 0.19 to 0.20 inches per year with no acceleration. RA-3 (maximum section) consistently shows the largest displacement (3.58 inches cumulative by 2025). Vertical settlement is less than 0.5 inches over 30 years and is not a concern. The constant rate confirms creep behavior — if a failure surface were developing, the rate would be expected to accelerate.

4.2 Arch Crest Plumbline Deflection

Table 4-2: Arch Crest Plumbline Deflection (+ = Downstream, Normalized to Sept.)

Year	PL-1 Left DS (mm)	PL-2 Crown DS (mm)	PL-3 Right DS (mm)	Pool (ft)	Concrete Temp (°F)
2015 (Sep)	+2.8	+4.5	+3.1	6,240.1	72
2016 (Sep)	+2.9	+4.6	+3.0	6,239.5	70
2018 (Sep)	+3.0	+4.8	+3.2	6,239.2	71
2020 (Sep)	+3.1	+5.0	+3.3	6,240.0	72
2022 (Sep)	+3.0	+5.2	+3.1	6,240.2	71
2025 (Sep)	+3.2	+5.5	+3.4	6,239.9	73

The arch crown plumbline (PL-2) shows a very gradual downstream drift of approximately 0.1 mm per year (1.0 mm total over 10 years). This is within the range attributed to ASR-induced expansion of the downstream face, which tends to push the crest downstream as the exterior concrete expands. The rate is slow and does not indicate structural distress, but should be monitored for acceleration as ASR progresses.

5.0 SUMMARY AND INTERPRETATIONS

Table 5-1: Ten-Year Monitoring Summary

Parameter	2015 Value	2025 Value	Change	Trend	Significance
Left abt. seepage	18.0 gpm	33.7 gpm	+87%	Increasing (~1.5 gpm/yr)	SIGNIFICANT — Investigate/remediate
Piezometer P-3	10.5 psi	18.0 psi	+71%	Increasing (~0.75 psi/yr)	SIGNIFICANT — Correlates w/ seepage
Right abt. disp.	2.01 in	3.07 in	+53%	Constant rate (0.20 in/yr)	MODERATE — Creep, monitor
Right abt. piez.	Stable	Stable	<1 ft	Stable	FAVORABLE — No pressure increase
Arch plumbline	+4.5 mm	+5.5 mm	+1.0 mm	Slow drift (0.1 mm/yr)	MINOR — ASR-induced
Right abt. seepage	7.2 gpm	7.2 gpm	0%	Stable	FAVORABLE
Arch gallery seep.	2.1 gpm	2.1 gpm	0%	Stable	FAVORABLE

The instrumentation data over the 10-year period from 2015 to 2025 provide a clear and consistent picture of dam behavior. The left abutment seepage increase is the most significant finding, with closely correlated piezometer P-3 data confirming the foundation seepage path identified by the 2025 boring program. The right abutment displacement continues at a constant rate with no indicators of instability. The arch concrete shows very slow ASR-induced expansion but no structural concern.

The dam is safe for continued operation. The recommended monitoring frequencies going forward are: weekly seepage measurements (left abutment), monthly piezometer readings (all), quarterly inclinometer and plumbline readings, and semi-annual survey monument readings. These frequencies should be maintained until the left abutment grouting program is completed and its effectiveness confirmed.